SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	CTRL CV
Project / Solution Name	CTRL CV
Name (Leader)	Muhammad Asynawi bin Azwarrudin
Email & Phone (Leader)	011-56482893 (asynawiazwarrudin@gmail.com)
University Affiliation(s)	Universiti Teknikal Malaysia Melaka
Live Prototype URL	https://ctrlcv-analyzeai.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/AbdulMuhaimin949/Aiversecity-Future-Skills

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The system architecture follows a linear pipeline beginning with a ‘profile' ingestion endpoint where Synthetic CVs are ingested. Data then enters a Data Cleansing & Normalization module for deduplication, UTM tracking removal, and non-career noise filtering via NLP regex. Standardized text is mapped against MASCO taxonomies and aligned with UN SDG goals. The core engine ('matchRoles' endpoint) uses a Deterministic Matching Algorithm executing Cosine and Skill-gap vector scoring. This backend integrates Live Real-Time Intelligence via the NewsData.io REST API and Anthropic task metrics. Outputs are sent to the frontend UI featuring Floating Widgets Radar and an AI chat guide.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.10+	Backend logic, NLP regex mapping, Cosine matching algorithm	PSF
NumPy / SciPy	Vector scoring and skill-gap array calculations	BSD
React / JavaScript	Frontend Floating UI Widgets, Animated Radar Scope	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
NewsData.io REST API	Live real-time intelligence for industry signals (zero drift)	Free / Tier 1
Anthropic API	Economic Index data, task augmentation metrics	Paid

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Local / Cloud File System	Persistent disk cache for newsdata_cache.json and safety evidence provenance	Standard

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
KrackDevs gpt-5.4-mini	KrackDevs	Pre-trained (Multilingual System Guide: EN, BM, Mandarin)	Proprietary
Anthropic 2024-03	Anthropic	Task augmentation & Scenario Uncertainty evaluation	Commercial API

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
MASCO 2020 & EMASCO	MOHR	Public (240 Occupations)	OpenGov	N
MYCOL 2024/2025	TalentCorp	Public	OpenGov	N
ESCO v1.2.1 Taxonomy	European Commission	Public (Skill hierarchy)	Open	N
Synthetic Student Profiles	Internal Script (Seed=42)	Synthetic	Internal	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

[/] Yes [ ] No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT	Brainstorming idea	moderate
Gemini	Search for real sources (API, dataset)	moderate
Claude	Codebase debug	major
Synthesia	Demo video development	moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The AI was strictly utilized to accelerate the coding of our frontend UI components and assist in syntax debugging, helping form a working product that strictly followed our self-thought pipeline and custom-designed architecture. All AI-generated code was manually reviewed, tested through unit testing against our MASCO taxonomy datasets, and thoroughly debugged by the team leader to ensure 100% adherence to our vector scoring logic and system design.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

[/] Yes [ ] No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Credits, platform	Used in hosting early stages of system development
Kracked Devs	API	Used in functioning LLM chatbox
SDC	Mentoring Session	Feedback to further improve development of the system

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

While highly convenient for the hackathon, both resources could be replaced if necessary. Replit credits were excellent for rapid prototyping and hosting early stages of development, but the system architecture allows migration to standard cloud instances like AWS or Heroku. Similarly, the Kracked Devs API effectively powered our LLM chatbox, but its function could be substituted with alternative LLM providers by adjusting the system prompts.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application actively filters non-career noise, removes UTM tracking, and enforces strict authorization and budget guardrails. All processed user data is duplicated immediately. No sensitive personal data is persistently sent to third-party AI APIs without anonymization.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

To mitigate hallucination, the system relies strictly on a Deterministic Matching Algorithm rather than generative matching. External APIs are caught locally to prevent rate limits and zero drift from external news counts. Scenario uncertainty is managed by locking projections to a baseline and a Fast Acceleration scenario.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

[/] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

[/] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

[/] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

[/] Any use of generative-AI tools for development has been disclosed and reviewed by the team.

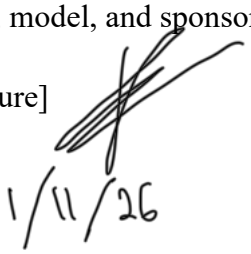
[/] Any use of official SDC sponsor resources has been declared in Section 5.

[/] The prototype described here is accessible to judges through the links provided in Section 1.

[/] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

[Date] 1/11/26